

Activity - 3

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1. Gradient Magnitude & Gradient Direction

Sol

Given

$$* G_x = 24$$

$$* G_y = 18$$

Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{24^2 + 18^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$= 30$$

Answer: Gradient Magnitude = 30

Gradient Direction

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$= \tan^{-1} \left(\frac{18}{24} \right)$$

$$= \tan^{-1}(0.75) \approx 36.87^\circ$$

Answer: Gradient Direction = $36.87^\circ (\approx 37^\circ)$

2. Non Maximum Suppression

Q. Given magnitude

Pixel	P_1	P_2	P_3	P_4	P_5
value	25	50	90	60	30

Gradient direction = Horizontal

Compare each pixel with its left and right neighbors

* $P_1 = \text{Border} \rightarrow 0$

* $P_2 : 50 < P_3 (90) \rightarrow \text{suppress}(0)$

* $P_3 : 90 > P_2 (50) \ \& \ 90 > P_4 (60) \rightarrow \text{keep}(90)$

* $P_4 : 60 < P_3 (90) \rightarrow \text{suppress}(0)$

* $P_5 = \text{Border} \rightarrow 0$

Answer

$$\left(\frac{0.90}{0.90} \right) \text{mag} = 0$$

Pixel	P_1	P_2	P_3, P_4	P_5
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After NMS 0 0 90 0 0

$$0.90 \cdot 0.90 = (0.90 \cdot 0) \text{mag} = 0$$

(F8) $0.90 \cdot 0.90 = 0.81$ (mag) = 0

3. Double Threshold Classification.

Sol High threshold = 100

Low threshold = 50

Pixel	Magnitude	Classification
P ₁	135	Strong Edge
P ₂	90	Weak Edge
P ₃	45	Non-Edge
P ₄	120	Strong Edge
P ₅	70	Weak edge
P ₆	20	Non-edge.

Answer:

- * strong edges : P₁, P₄
- * weak edges : P₂, P₅
- * Non-edges : P₃, P₆

4. Hysteresis Thresholding

sol Given

Pixel	P_1	P_2	P_3	P_4	P_5	P_6	P_7
Magnitude	130	95	40	115	65	25	105

High threshold = 100

Low threshold = 50

Step 1: Strong edges (≥ 100)

* $P_1 = 130$

* $P_4 = 115$

* $P_7 = 105$

Strong edges: P_1, P_4, P_7

Step 2: Weak edges (50-99)

* $P_2 = 95$

* $P_5 = 65$

Weak edges: P_2, P_5

Step 3: Non-Edges (< 50)

* $P_3 = 40$

* $P_6 = 25$

N
Step 4 : Hysteresis

Given :

- * P_2 is connected to $P_1 \rightarrow$ keep
- * P_5 is not connected to any strong edge $\rightarrow$ discard.

